

AS Level Physics (Edexcel IAL) — Formula Sheet

Edexcel International AS: Units 1–2 (WPH11/12). The exam gives the data booklet; know what EVERY symbol means and when each applies.

UNIT 1 — MECHANICS & MATERIALS

Speed/velocity	$v = \frac{s}{t}$; average velocity = displacement/time
Acceleration	$a = \frac{v}{t}$ (m/s^2)
suvat	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$
Force	$F = ma$ (N)
Weight	$W = mg$ $g = 9.81 \text{ N/kg}$ on Earth
Momentum	$p = mv$; conservation: $\sum p \text{ before} = \sum p \text{ after}$
Moments	moment = $F \times \text{perpendicular distance}$
Work & energy	$W = F \cdot s \cdot \cos \theta$; $KE = \frac{1}{2}mv^2$; $GPE = mgh$
Power	$P = E/t = W/t$ (W)
Efficiency	= useful energy out / total energy in
Stokes' law	$F = 6\pi \eta r v$ (viscous drag on sphere)
Hooke's law	$F = kx$ k : stiffness (N/m)
Stress & strain	stress $\sigma = F/A$ (Pa); strain $\epsilon = x/x$ (no units)
Young modulus	$E = \sigma/\epsilon$ (Pa) — gradient of stress-strain graph
Elastic strain energy	$E = \frac{1}{2}Fx = \frac{1}{2}kx^2$ (area under F - x graph)
Density	$\rho = m/V$ (kg/m^3)
Upthrust	= weight of displaced fluid (Archimedes)

UNIT 2 — WAVES & ELECTRICITY

Wave speed	$v = f\lambda$
Period & frequency	$T = 1/f$
Refractive index	$n = \frac{\sin i}{\sin r} = \frac{c}{v}$; $n_1 \sin \theta_1 = n_2 \sin \theta_2$ (Snell)
Critical angle	$\sin C = 1/n$ (denser to less dense)
Diffraction grating	$n\lambda = d \sin \theta$
Photoelectric	$hf = \phi + \frac{1}{2}mv^2_{\text{max}}$; $hf \geq \phi$ (threshold)
Photon energy	$E = hf = hc/\lambda$
de Broglie	$\lambda = h/p = h/mv$
Current	$I = \frac{Q}{t}$ (A, C, s)
Potential difference	$V = W/Q$ (J/C)
Resistance	$R = V/I$ (Ω)
Ohm's law (conductors)	$I \propto V$ at constant temperature
Resistivity	$R = \rho L/A$ ρ : ($\Omega \cdot \text{m}$)
Power	$P = VI = I^2R = V^2/R$
Series/parallel R	series: add; parallel: $1/R = 1/R_1 + 1/R_2 + \dots$
EMF & internal resistance	$\epsilon = I(R + r)$; terminal $V = \epsilon - Ir$
Potential divider	$V_{\text{out}} = V_{\text{in}} \times \frac{R_2}{R_1 + R_2}$
Drift velocity	$I = nqvA$ n : carriers/ m^3 , v : drift speed

CONSTANTS (given — know typical values)

c, g, h, e	$3.00 \times 10^8 \text{ m/s}$; 9.81 N/kg ; $6.63 \times 10^{-34} \text{ J}\cdot\text{s}$; $1.60 \times 10^{-19} \text{ C}$
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